The roots of the general octic equation

Soumadeep Ghosh

Kolkata, India

Abstract

In this paper, I describe the roots of the general septic equation.
The paper ends with ”The End”

Introduction

In a previous paper, Ive described how to solve the general quintic equation.

In a previous paper, Ive described how to solve the general sextic equation.

In a previous paper, Ive described how to solve the general septic equation.

In a previous paper, Ive described my monic octic identity.

In a previous paper, Ive described how the roots of my monic octic equation are expressible in radicals.

In this paper, I describe the roots of the general octic equation.

Preliminaries

The general octic equation is

$$ax^8 + bx^7 + cx^6 + dx^5 + ex^4 + fx^3 + gx^2 + hx + i = 0$$

where a,b,c,d,e,f,g,h,i are constants.

If $i = 0$ then the equation reduces to

$$x(ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h) = 0$$

which has the root $x = 0$ and the septic

$$ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h = 0$$

which can be solved.

Similarly, if $a = 0$ then the equation reduces to septic equation

$$bx^7 + cx^6 + dx^5 + ex^4 + fx^3 + gx^2 + hx + i = 0$$

which can be solved.

If $a = i = 0$ then the equation reduces to

$$x(bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h) = 0$$

which has the root $x = 0$ and the sextic

$$bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h = 0$$

whose roots are known.

Therefore, $a \neq 0$ and $i \neq 0$ henceforth. Moreover, we divide the general octic equation by the leading coefficient a to transform the general octic equation to the monic octic equation

$$x^8 + ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h = 0$$

where $h \neq 0$ henceforth.

Comparing coefficients with my monic octic

Recall my monic octic.

Comparing coefficients, we get

1. $a = a$
2. $b = (a - P)P + \frac{f-g}{Q} + Q$
3. $c = \frac{(a-P)(f-g)+g-h}{Q} + PQ + Q$
4. $d = f - g + \frac{(a-P)(g-h)+h}{Q} + PQ + Q$
5. $e = \frac{ah-h(P+Q)}{Q} + f + PQ$
6. $f = f$
7. $g = g$
8. $h = h$

Thus, if suitable P and Q are obtained, then, by my monic octic identity, we can reduce the monic octic equation to the product of a quartic equation and a quartic equation, whose roots are known.

Choosing P and Q

Eliminating b, c, d between equations (2), (3), (4) and (5) gives us the eliminant

$$e = \frac{ah}{Q} + f - \frac{hP}{Q} - h + PQ$$

which can be inverted to give

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}$$

As long as $Q(h - Q^2) \neq 0$ and we obtain corresponding P and Q, we may choose any value for Q to obtain P. For most octics, Q = a is a valid and convenient choice. When Q = a is not a valid choice, other valid and convenient choices may be $Q = \frac{h}{2}$, $Q = P$ etc.

Solving the monic octic

Once we have at least one valid value of P and one valid value of Q, by the right side of my monic octic identity, we obtain 8 roots of the monic octic equation.

Notes

1. Note that by following this procedure, we obtain 8 roots of the general octic equation expressible in radicals.
2. Note that this procedure doesnt invalidate Galois theory since our procedure is based on expressing the general octic equation as factored forms but not on solving the octic equation algebraically.

Exercises for the reader

Find the roots of the following octic equations expressible in radicals:

1. $$2x^8 + 24x^7 - 204x^6 + 50x^5 - 156x^4 + 3460x^3 - 354x^2 + 34x + 600 = 0$$

2. $$x^8 - 27x^3 + 14x^2 + 25 = 0$$

The End